

D287

Java Frameworks

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A reporting module for a hospital supply chain calls `Files.readAllLines(path)`, which declares throws `IOException`. The developer, who joined the team last month, wraps nothing in `try` and adds no `throws` clause. The file is only 2 KB. What happens?

- A. It compiles because the file is small enough to read safely.
- B. It compiles and crashes only if the file is missing.
- C. The code does not compile until `IOException` is caught or declared.
- D. It compiles because every exception is optional to handle.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which exception can a method throw without catching it and without listing it in a `throws` clause?

- A. `IOException`
- B. `Exception`
- C. `IllegalArgumentException`
- D. `SQLException`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An order service used by 50 warehouse staff has `try { ... } catch (Exception e) { ... } catch (NumberFormatException e) { ... }`. The lead wants both messages logged in the new format. What is wrong?

- A. `NumberFormatException` is checked and needs a `throws` clause.
- B. Nothing; Java checks every `catch` block and runs the best match.
- C. Both catches run, so the message is logged twice.
- D. The second catch is unreachable, so the code does not compile; put `NumberFormatException` first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A method has `try { return 1; } finally { System.out.println(done); }` where `done` is a `String` variable. What happens when the method is called?

- A.** It returns 1 and nothing prints, because `return` exits immediately.
- B.** It prints only if an exception occurs.
- C.** It does not compile, because `finally` cannot follow a `return`.
- D.** It prints the value of `done` and then returns 1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A loan application built for a credit union with 12 branches must stop processing when an applicant's income is negative. The developer creates class `InvalidIncomeException` extends `Exception`. Inside `validate()`, what raises it?

- A.** `return new InvalidIncomeException();`
- B.** `throws new InvalidIncomeException();`
- C.** `catch (InvalidIncomeException e) {}`
- D.** `throw new InvalidIncomeException(msg);`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a Spring Boot inventory app deployed to three regional data centers, a controller needs a `ProductRepository`. The team prefers constructor injection. How should the controller get the repository?

- A.** Declare `ProductRepository` as a constructor parameter and let the framework supply it.
- B.** Create one repository per data center in the controller.
- C.** Make the repository a static field that each controller fills in at startup.
- D.** Call `new ProductRepository()` inside the controller.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A parts store's web page, styled in the company's blue brand colors, lets staff submit a new part. A rule says price must exceed cost. Where does the price rule belong?

- A. In the model or service layer.
- B. In the view template.
- C. In the controller only.
- D. In the CSS file with the brand colors.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

In an MVC web app, which component receives the HTTP request for a form submission and decides which page to show next?

- A. The controller.
- B. The view.
- C. The database.
- D. The model.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A Drone class in a delivery startup's fleet software must reuse code from the Vehicle class and also meet the Trackable and Chargeable contracts. The fleet has 200 drones. Which declaration is legal?

- A. class Drone extends Vehicle, Trackable, Chargeable
- B. class Drone extends Vehicle implements Trackable, Chargeable
- C. class Drone implements Vehicle extends Trackable, Chargeable
- D. No declaration works; a class can use only one parent type.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A conference app for a 1,500-person trade show stores badge numbers already scanned at the door. Each badge may be counted only once, and order does not matter. The venue has four entrances. Which collection fits best?

- A. `HashMap<Integer, Integer>`
- B. `Integer[]` of size 4
- C. `HashSet<Integer>`
- D. `ArrayList<Integer>`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A program must look up a customer's name by a unique customer ID. Which collection fits best?

- A. `Set<Integer>`
- B. `Map<Integer, String>`
- C. `List<String>`
- D. `Set<String>`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A booking form built with a popular JavaScript library already blocks guest counts above 10. The back end, written in Java, saves the booking directly. Marketing wants the form released next week. Should the back end also check the guest count?

- A. No; the form check already makes invalid data impossible.
- B. Yes; requests can reach the server without the form, so it must validate.
- C. No; the release date leaves no time for a second check.
- D. Only the database needs a check, not the Java code.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A reporting module ~~for a hospital supply chain~~ calls `Files.readAllLines(path)`, which declares throws `IOException`. The developer, ~~who joined the team last month~~, wraps nothing in try and adds no throws clause. ~~The file is only 2 KB.~~ What happens?

Target word: effect of an unhandled checked exception **Decides:** `IOException` is checked, so it must be caught or declared.

A	Camouflage	File size has no effect on compile-time exception rules.
B	Confusion set	Describes an unchecked exception; <code>IOException</code> is checked.
C	Correct	Checked exceptions are enforced by the compiler.
D	Absolute language	Only unchecked exceptions are optional.

2. Answer C Concept 1 · transfer

Which exception can a method throw without catching it and without listing it in a throws clause?

Target word: identify an unchecked exception **Decides:** `RuntimeException` subclasses are unchecked.

A	Inverted operation	Checked; must be caught or declared.
B	Adjacent category	The parent of checked exceptions; throwing it must be declared.
C	Correct	It extends <code>RuntimeException</code> , so the compiler does not require handling.
D	Confusion set	Also checked, like <code>IOException</code> .

3. Answer D Concept 2

An order service ~~used by 50 warehouse staff~~ has `try { ... } catch (Exception e) { ... } catch (NumberFormatException e) { ... }`. ~~The lead wants both messages logged in the new format.~~ What is wrong?

Target word: problem with the catch order **Decides:** The general `Exception` catch comes before the more specific one.

A	Adjacent category	It is unchecked; the issue is ordering.
B	Confusion set	Java takes the first matching catch, top to bottom.
C	Inverted operation	At most one catch block runs per exception.
D	Correct	A subclass catch after its parent can never run, which is a compile error.

4. Answer D Concept 2 · transfer

A method has `try { return 1; } finally { System.out.println(done); }` where `done` is a `String` variable. What happens when the method is called?

Target word: whether finally runs after a return **Decides:** finally runs before control leaves the method.

A	Inverted operation	Return waits for finally to finish.
B	Confusion set	That describes catch, not finally.
C	Absolute language	try with return and finally is legal.
D	Correct	finally executes even when the try block returns.

5. Answer D Concept 3

A loan application ~~built for a credit union with 12 branches~~ must stop processing when an applicant's income is negative. The developer creates class `InvalidIncomeException` extends `Exception`. Inside `validate()`, what raises it?

Target word: the statement that raises an exception **Decides:** throw raises a specific exception object inside the method body.

A	Adjacent category	Returns an object; nothing is raised.
B	Confusion set	throws belongs in the method signature, not as a statement.
C	Inverted operation	Handles the exception instead of raising it.
D	Correct	throw plus a new exception object raises it; the method must also declare throws because it is checked.

6. Answer A Concept 4

In a Spring Boot inventory app ~~deployed to three regional data centers~~, a controller needs a `ProductRepository`. The team prefers constructor injection. How should the controller get the repository?

Target word: how a framework-managed class gets a dependency **Decides:** The framework creates the bean and passes it in; the class declares it as a constructor parameter.

A	Correct	That is dependency injection: the container builds and wires the bean.
B	Camouflage	Deployment count does not decide how dependencies are wired.
C	Adjacent category	Shared static state bypasses the container.
D	Inverted operation	The class builds its own dependency, the opposite of inversion of control.

7. Answer A Concept 5

A parts store's web page, ~~styled in the company's blue brand colors~~, lets staff submit a new part. A rule says price must exceed cost. Where does the price rule belong?

Target word: which MVC layer owns a business rule **Decides:** Business rules belong in the model (or service layer behind it).

A	Correct	The model holds data and the rules that govern it.
B	Confusion set	The view displays data; it should not enforce business rules.
C	Adjacent category	The controller routes the request and calls the model; putting rules there spreads them across endpoints.
D	Camouflage	Styling has nothing to do with business logic.

8. Answer A Concept 5 • transfer

In an MVC web app, which component receives the HTTP request for a form submission and decides which page to show next?

Target word: role of the controller **Decides:** Handling the request and choosing the view is the controller's job.

A	Correct	It maps the request, calls the model, and returns a view.
B	Confusion set	The view renders output; it does not route requests.
C	Adjacent category	Storage sits behind the model, outside MVC's request flow.
D	Inverted operation	The model holds data and rules; it does not handle requests.

9. Answer B Concept 6

A Drone class in a delivery startup's fleet software must reuse code from the Vehicle class and also meet the Trackable and Chargeable contracts. ~~The fleet has 200 drones.~~ Which declaration is legal?

Target word: legal inheritance combination **Decides:** One class may be extended; many interfaces may be implemented.

A	Confusion set	Java does not allow extending more than one type with a class.
B	Correct	Single class inheritance plus multiple interfaces is allowed.
C	Inverted operation	Swaps the keywords; extends comes first and names the class.
D	Absolute language	Multiple interfaces are allowed.

10. Answer C Concept 7

A conference app ~~for a 1,500-person trade show~~ stores badge numbers already scanned at the door. Each badge may be counted only once, and order does not matter. ~~The venue has four entrances.~~ Which collection fits best?

Target word: collection that enforces uniqueness **Decides:** Only once, order does not matter.

A	Confusion set	A map is for key-value lookups; no value is needed here.
B	Wrong stem number	Sizes the store to the entrance count, not the badges.
C	Correct	A Set rejects duplicates, with fast membership checks.
D	Adjacent category	Lists allow duplicates and need extra checks.

11. Answer B Concept 7 · transfer

A program must look up a customer's name by a unique customer ID. Which collection fits best?

Target word: collection for keyed lookup **Decides:** Lookup by unique key is a Map.

A	Dropped element	Keeps the IDs but drops the names.
B	Correct	Maps store unique keys with values for direct lookup.
C	Adjacent category	Index is position, not customer ID.
D	Inverted operation	Stores names with no key to look them up by.

12. Answer B Concept 8

A booking form ~~built with a popular JavaScript library~~ already blocks guest counts above 10. The back end, written in Java, saves the booking directly. ~~Marketing wants the form released next week.~~ Should the back end also check the guest count?

Target word: need for server-side validation **Decides:** Client checks can be bypassed by direct requests.

A	Absolute language	A user can send a request that skips the form.
B	Correct	Server-side validation is the real guard; the UI check is a convenience.
C	Camouflage	Schedule does not change whether validation is required.
D	Adjacent category	A database constraint helps, but the application should reject bad input with a clear error.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	1	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Checked versus unchecked exceptions	Checked exceptions (Exception subclasses outside RuntimeException) must be caught or declared with throws; unchecked (RuntimeException and its subclasses) need not be.	Thinking every exception must be caught, or that NullPointerException is checked.	A method calls something that throws IOException or SQLException without a try or throws.
2	try, catch order and finally	Catch blocks are tested top to bottom, so specific types go before general ones; finally runs whether or not an exception occurs.	Putting catch (Exception) first, or thinking finally is skipped after a return or exception.	A stack of catch blocks or a finally block with a print statement.
3	throw versus throws and custom exceptions	throw creates and raises one exception object inside a method; throws in the signature declares what a method may pass up; a custom exception extends Exception or RuntimeException.	Swapping throw and throws, or extending Error for business rules.	A business rule violation needs its own exception type.
4	Inversion of control and dependency injection	The framework creates and wires objects (beans) and hands dependencies to a class, instead of the class building them with new.	Calling new on a service inside a controller managed by the framework.	A class needs a repository or service and the question asks how it gets one.
5	Model-View-Controller roles	The model holds data and rules, the view displays it, and the controller takes the request, calls the model, and picks the view.	Putting business logic or database calls in the view template.	Which layer should handle a form submission, a price rule, or page layout.
6	Interfaces versus abstract classes	A class can implement many interfaces but extend only one class; abstract classes can hold state and constructors, interfaces define a contract.	Believing Java allows extending two classes.	A class needs behavior from two sources.
7	Choosing a collection	List keeps order and allows duplicates; Set forbids duplicates; Map stores key-value pairs with unique keys.	Using a List when duplicates must be blocked or a lookup by key is needed.	"No duplicates" or "look up by ID" in the requirement.
8	Server-side validation	Validate input on the server even when the user interface validates too, because client checks can be bypassed.	Trusting a front-end form check as the only guard.	The UI already blocks bad values and the question asks whether the back end must check.

Course scope (public description): Java Frameworks builds object-oriented programming expertise and introduces powerful new tools for Java application development. Students will execute exception handling, Java frameworks, and other object-oriented principles and constructs to develop a complete application including a user interface. This course requires foundational knowledge of object-oriented programming and the Java language.